

Recall — angles

Naming, measuring and estimating angles before the calculating begins.

LESSON 80

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 5 ENTRY

LESSON CLOCK

10'

12'

12'

6'

Naming	10 min
Measuring	12 min
Estimating	12 min
Naming with three letters	6 min

LEARNING OBJECTIVES

- Name angles as acute, right, obtuse, straight or reflex.
- Measure and draw an angle with a protractor.
- Estimate an angle before measuring it.
- Use the three-letter naming of an angle.

TEXTBOOK

National	pp. 223–226
Cambridge	Stage 7, pp. 55–58
Workbook	Exercise 5.1

01

Explanation

Naming

NAME	SIZE	EVERYDAY EXAMPLE
acute	between 0° and 90°	a slice of pizza
right	90°	the corner of a book
obtuse	between 90° and 180°	a laptop half open
straight	180°	a flat line
reflex	between 180° and 360°	the outside of a corner
full turn	360°	once round

A RIGHT ANGLE IS THE RULER FOR THE OTHERS

Acute is less than a corner, obtuse is more. Holding up a corner of paper is a perfectly good way to classify an angle before measuring it.

Measuring

STEP	WHAT TO DO
1	put the centre of the protractor on the vertex
2	lay the base line along one arm
3	read where the other arm crosses the scale
4	choose the scale that agrees with your estimate

TWO SCALES, ONE RIGHT ANSWER

A protractor reads 50° and 130° at the same place. Estimating first — acute or obtuse? — decides which to take, and it is why step 4 exists.

Estimating

LANDMARK	SIZE	RECOGNISED BY
a right angle	90°	a square corner
half of it	45°	a diagonal of a square
a third of it	30°	a thin wedge
two right angles	180°	a straight line
a little over a right angle	100°	just past square

ESTIMATE TO THE NEAREST LANDMARK FIRST

“A bit more than 90° ” is enough to reject a reading of 80° . The estimate is not the answer; it is the check on the answer.

Naming with three letters

$\angle ABC$ means the angle at B , between the arms BA and BC . The middle letter is always the vertex.

WRITTEN	VERTEX	ARMS
$\angle ABC$	B	BA and BC
$\angle XYZ$	Y	YX and YZ
$\angle B$	B	only if there is no ambiguity

THE MIDDLE LETTER IS THE VERTEX

$\angle ABC$ and $\angle CBA$ are the same angle; $\angle BAC$ is a different one. Reading the middle letter first is the habit to build.

02 Worked examples

EXAMPLE 1 Classify angles of 42° , 90° , 137° and 210° .

- 1 42° — acute.
- 2 90° — right.
- 3 137° — obtuse.
- 4 210° — reflex.

ANSWER

Acute, right, obtuse, reflex

EXAMPLE 2 A protractor reads either 50° or 130° . The angle looks clearly less than a corner. Which is it?

① Less than a right angle means acute.

② Acute means under 90° .

③ 50°

ANSWER

50°

EXAMPLE 3 In $\angle PQR$, which point is the vertex?

① The middle letter names the vertex.

② Q.
The arms are QP and QR .

ANSWER

Q

03 Interactive model

Have the class estimate ten angles on the board before any protractor comes out; measurement errors of a hundred degrees disappear once estimating is a habit.

Name that angle

42° is:

acute

right

obtuse

reflex

137° is:

acute

right

obtuse

reflex

210° is:

acute

obtuse

straight

reflex

180° is:

obtuse

straight

reflex

a full turn

A full turn is:

180°

270°

360°

90°

In $\angle ABC$ the vertex is:

A

B

C

any of them

$\angle ABC$ and $\angle CBA$ are:

the same angle

different angles

supplementary

complementary

A protractor showing 50 and 130 for an obtuse angle means:

50°

130°

either

neither

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Angle	Burchak	Угол
Vertex	Uchi	Вершина
Arm	Tomon	Сторона
Acute angle	O‘tkir burchak	Острый угол
Right angle	To‘g‘ri burchak	Прямой угол
Obtuse angle	O‘tmas burchak	Тупой угол
Reflex angle	Yoyiq burchakdan katta	Развёрнутый и больше
Protractor	Transportir	Транспортир

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An acute angle is:

under 90°

over 90°

exactly 90°

over 180°

An obtuse angle is:

under 90°

between 90° and 180°

over 180°

exactly 180°

A reflex angle is:

under 90°

between 90° and 180°

between 180° and 360°

exactly 360°

The protractor is placed with its centre on:

an arm

the vertex

the page

the scale

Estimating before measuring decides:

the arms

which scale to read

the vertex

the units

In $\angle XYZ$ the arms are:

XY and XZ

YX and YZ

ZX and ZY

none of these

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Classify 42°

ANSWER Acute

2. Classify 90°

ANSWER Right

3. Classify 137°

ANSWER Obtuse

4. Classify 180°

ANSWER Straight

5. Classify 210°

ANSWER Reflex

6. A full turn

ANSWER 360°

7. The vertex of $\angle ABC$

ANSWER B

Medium · the standard question

7 PROBLEMS

1. Classify 89° and 91°

ANSWER Acute and obtuse

2. A protractor reads 50 or 130 for an acute angle

ANSWER 50°

3. Half a right angle

ANSWER 45°

4. A third of a right angle

ANSWER 30°

5. Two right angles

ANSWER 180°

6. Three right angles

ANSWER 270°

7. Are $\angle ABC$ and $\angle CBA$ the same?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. The reflex angle beside an angle of 130°

ANSWER 230°

2. The reflex angle beside a right angle

ANSWER 270°

3. An angle and its reflex partner add to

ANSWER 360°

4. Why does a protractor carry two scales?

ANSWER So it can be laid along either arm

5. The smallest reflex angle

ANSWER Just over 180°

6. Estimate the angle between the hands at 3 o'clock

ANSWER 90°

7. And at 6 o'clock

ANSWER 180°

07 Homework

Homework — 5 tasks

Estimate every angle before you measure it, and write both numbers down.

1. Classify angles of 37° , 90° , 154° and 265° .
2. Draw angles of 35° , 115° and 90° with a protractor.
3. Measure four angles from your textbook, estimating each one first.
4. Name the vertex and the arms of $\angle MNP$.
5. Find the reflex angle that goes with an angle of 145° .